The Journal of Open Source Software

DUGseis: A Python package for real-time and post-processing of picoseismicity

Martina Rosskopf [1,2¶], **Virginie Durand** [3], **Linus Villiger** [2], **Joseph Doetsch** [4], **Anne Obermann** [1,2], and **Lion Krischer** [5]

1 Institute of Geophysics, ETH Zurich, Zurich, Switzerland **2** Swiss Seismological Service, ETH Zurich, Zurich, Switzerland **3** GeoAzur, Université Côte d'Azur, France **4** Lufthansa Industry Solutions, Raunheim, Germany **5** Mondaic AG, Zurich, Switzerland ¶ Corresponding author

DOI: 10.xxxxxx/draft

Software

- Review
- Repository
- Archive

Editor: Chris Vernon

Reviewers:

- @trichter
- @erexer
- @jessepisel

Submitted: 09 February 2024
Published: unpublished

License
Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License (CC BY 4.0).

DRAFT

Summary

Detecting earthquakes and compiling them to earthquake catalogs are fundamental tasks in seismology. Acoustic emission sensors allow the detection of tiniest so called picoseismic events, representing fractures on the cm or dm scale (-5<magnitude<0), even smaller seismic events are recorded in the laboratory. Picoseismic events have corner frequencies of 1 kHz-100 kHz. They are typically recorded with sampling frequencies of 200 kHz to 1 MHz and cannot be handled by standard seismic processing software that is developed for signals with frequencies f<500 Hz. Commercial software for continuous recording and processing of picoseismicity exists (e.g. GMuG-conti, fastloc.REEL) and is successfully used in scientific in-situ experiments (e.g. hydraulic stimulation, earthquake nucleation, nuclear waste disposal). However, commercial software comes with restrictions unfavorable for university-based in-situ experiments in underground laboratories, where tailoring the software to experiment specific questions or adding newly developed analysis techniques are routine operations in order to advance the study of the rock response in great detail. The open-source DUGseis software package fills this gap. It has been developed to manage, process and visualize continuous, high-frequency seismic data. The package can be used to create real-time and post-processing earthquake catalogs and to directly visualize their event waveforms and locations in a graphical interface. As the software is Python-based, users can easily add their own processing routines.

Statement of need

The open-source, Python-based DUGseis package is designed to align with the functionalities of SeisComP (Helmholtz-Centre Potsdam - GFZ German Research Centre for Geosciences and gempa GmbH, 2008), a standard software used in microseismic large-scale earthquake processing (<500Hz, M>-0.5). DUGseis is tailored for picoseismic events (-5<M<0) with much higher frequency ranges between 1 kHz - 100 kHz (Plenkers et al., 2022), as recorded by acoustic emission sensors (AE sensors). High-frequency seismic data processing is common in mining environments to monitor tunnel stability and has become very popular in underground laboratories (Plenkers et al., 2022). To date, many projects used commercial software (e.g. GMuG-conti from GMuG Gesellschaft für Materialprüfung und Geophysik, Bad Nauheim/Germany, www.gmug.eu or fastloc.REEL from Grzegorz Kwiatek, www.induced.pl/software/fastloc-reel) lacking the flexibility to adapt to the specific needs of the project since they are not open-source. Additionally, many projects used trigger-based recordings, which means that the incoming waveform data is only saved to disk when a pre-set trigger threshold is reached by a recorded event. A disadvantage of the triggered recording strategy is the so-called dead time. Once an event has been triggered, no further event can be triggered until the processing of the first triggered event has been completed. The dead time can be a multiple of the recording time, meaning

that important events can be missed if another event has just occurred before. Eliminating this dead time is important when high event rates are expected. With DUGseis it is possible to record and store (Linus Villiger et al., 2024) continuous waveform data in the MHz range and process the data directly, removing these dead times. As DUGseis is open-source and Python-based, researchers have the flexibility to complement the processing with their own Python-based codes, adapted to the need of their project. Compared to commercial software this flexibility of DUGseis is a major advantage.

Functionality and Features

The DUGseis software is a Python-based package that focuses on processing continuous high-frequency data, extracting waveforms of picoseismic events and creating an earthquake catalog. To make its use and output more accessible to seismologists, some features use modules and functions from the ObsPy package (Beyreuther et al., 2010), a popular package in seismology.

The DUGseis software requires sensor metadata with sensor locations as StationXML files and continuous waveform data in the ASDF format (D'Avella et al., 2023; Greenfield et al., 2015) to run. Here, the continuous waveform data was acquired using DUGseis acquisition (Linus Villiger et al., 2024), another software package developed specifically for the Bedretto Underground Laboratory to record and store waveform data from AE sensors in the ASDF format using specific Spectrum digitizer cards. For general use of the DUGseis package presented here, the waveform data does not need to be acquired from the DUGseis acquisition, but the ASDF format can be transferred from other data formats.

DUGseis retrieves all important information regarding data directories and processing settings from a configuration file. The configuration file can be used to open a graphical interface or passed to the processing script. After processing the continuous waveform data, DUGseis outputs an event catalog as a database, which can also be saved as QuakeML or CSV files. Figure 1 shows the input and output of the DUGseis processing.

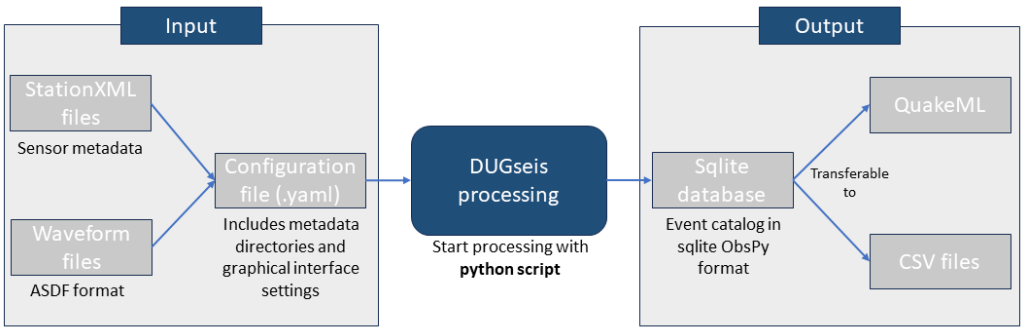

Figure 1: In- and output of DUGseis processing.

Data processing

The processing itself is done within Python scripts that call various DUGseis functions, increasing flexibility of use.

DUGseis was mainly developed to process continuous waveform data recorded by AE sensors. L. Villiger et al. (2020) used an initial version of the DUGseis software at the Grimsel Laboratory. The software was further developed for the hydraulic stimulation experiments performed at the Bedretto Underground Laboratory (Obermann et al., 2024). For this purpose, it was necessary to process data in real-time during the stimulations in order to assess the hazard potential of the ongoing stimulation and to gain a direct understanding of the fluid propagation in the rock volume. In addition, more detailed post-processing was required after stimulation to learn more about the rock volume response. Both, real-time analysis and post-processing can be

carried out using the DUGseis package. During the live processing, directories are monitored in real time for new incoming data. For the post-processing, all directories containing stored continuous waveform data are passed to the processing script. This continuous data is then re-processed during replay, which is a major advantage over trigger-based software, where only the waveforms of the already triggered events could be revisited.

The DUGseis package includes many typical seismological processing steps to produce earthquake catalogs. The steps may include a detection stage to select event candidates on a number of predefined sensors. This step can be useful to speed up the processing, especially in real-time. Other processing steps are the picking of all traces with different pickers (e.g. STA/LTA), the localization of events with a basic location algorithm and the estimation of magnitudes, currently based on acoustic emission sensors. Several methods are implemented for some of the processing steps, whcih can be selected by the user. Furthermore, all processing steps can be adapted to the needs of the researcher. For more detailed information on which steps were used in the Bedretto Underground laboratory, see Obermann et al. (2024).

After processing, the output of the events, including arrivals and picks, are stored in a database and can also be saved as QuakeML or CSV files.

Graphical Interface

Another feature of DUGSeis is its ability to visualize the recorded waveform data and to allow manual repicking and relocating, both in real-time and in post-processing. The graphical interface (GUI) displays not only waveforms, but also the output of the processing stage, such as the origin and pick times of an event. Additionally, the event and sensor locations are shown in a 3D visualization. Figure 2 shows the layout of the graphical user interface.

The GUI provides the ability to inspect each event and view the channels that recorded the event. In addition, manual repicking and relocating can be done here.

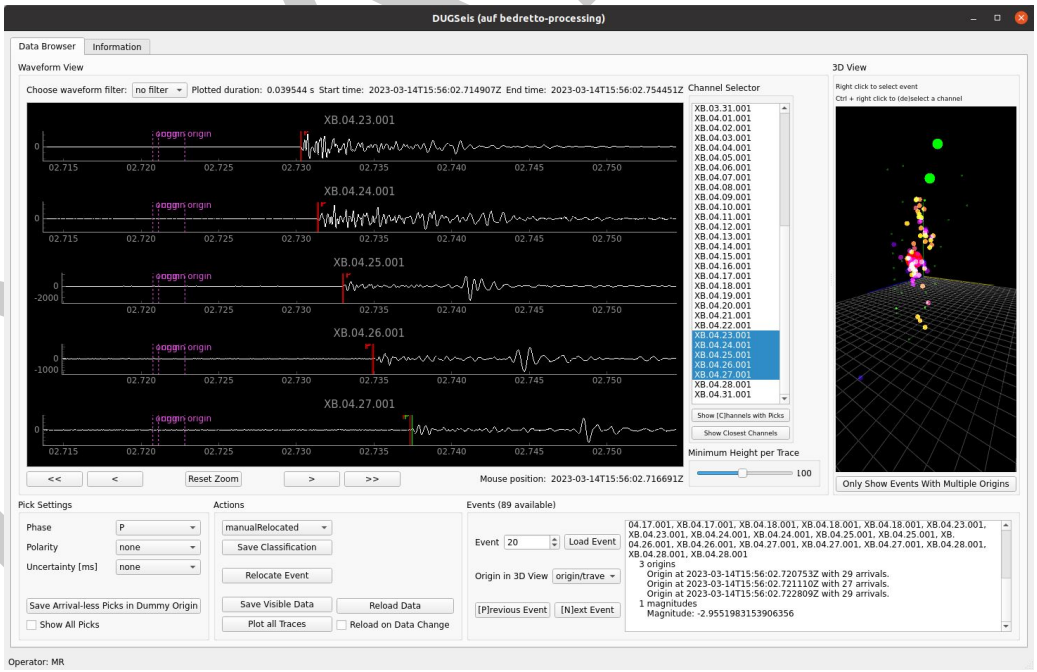

Figure 2: Graphical interface of GUI with waveforms and picks of an event and the 3D plot of the events in the database. Data from Rosskopf et al. (2024).

Usage

Since 2021 dozens of hydraulic stimulation experiments have been performed at the Bedretto Underground Laboratory in Switzerland (Ma et al., 2022; Obermann et al., 2024; Plenkers

et al., 2023). In this context, the DUGseis package was used to detect picoseismicity by processing the incoming high-frequency waveform data in real-time and in post-processing mode (Obermann et al., 2024). A small seismic dataset from one of the hydraulic stimulations can be found in (Rosskopf et al., 2024).

Acknowledgements

The submitting author, Martina Rosskopf, is funded by SNF Project "Characterizing and understanding Enhanced Geothermal Systems (EGS) - novel tools and applications in a deep underground laboratory" (200021_192151). The BedrettoLab is financed by ETH Zürich and the Werner Siemens Foundation. This paper is BULGG publication BPN_023.

References

Beyreuther, M., Barsch, R., Krischer, L., Megies, T., Behr, Y., & Wassermann, J. (2010). ObsPy: A python toolbox for seismology. *Seismological Research Letters*, *81*, 530–533. https://doi.org/10.1785/gssrl.81.3.530

D'Avella, D., Graham, B., Jamieson, W., Droettboom, M., Slavich, E., Robitaille, T., Dencheva, N., Greenfield, P., Simon, B., MacDonald, K., Bray, E. M., Burnett, Z., Davies, J., Mumford, S., Markovtsev, V., Tollerud, E., Sipőcz, B., Bradley, L., Fabry, Ç., … Kerzendorf, W. (2023). *Asdf-format/asdf: 3.0.1*. Zenodo. https://doi.org/10.5281/zenodo.10054611

Greenfield, P., Droettboom, M., & Bray, E. (2015). ASDF: A new data format for astronomy. *Astronomy and Computing*, *12*, 240–251. https://doi.org/10.1016/j.ascom.2015.06.004

Helmholtz-Centre Potsdam - GFZ German Research Centre for Geosciences and gempa GmbH. (2008). *The SeisComP seismological software package. GFZ Data Services.* https://doi.org/10.5880/GFZ.2.4.2020.003

Ma, X., Hertrich, M., Amann, F., Bröker, K., Doonechaly, N. G., Gischig, V., Hochreutener, R., Kästli, P., Krietsch, H., Marti, M., Nägeli, B., Nejati, M., Obermann, A., Plenkers, K., Rinaldi, A. P., Shakas, A., Villiger, L., Wenning, Q., Zappone, A., … Giardini, D. (2022). Multi-disciplinary characterizations of the BedrettoLab-a new underground geoscience research facility. *Solid Earth*, *13*, 301–322. https://doi.org/10.5194/se-13-301-2022

Obermann, A., Rosskopf, M., Durand, V., Plenkers, K., Bröker, K., Doonechaly, N. G., Gischig, V., Hertrich, M., Kästli, P., Ma, X., Hansruedi, M., Rinaldi, A. P., Villiger, L., Wiemer, S., Zappone, A., & Giardini, D. (2024). Picoseismic response of hectometer-scale fracture systems to stimulation with cm-scale resolution under the Swiss Alps, in the Bedretto Underground laboratory. *Journal of Geophysical Research (in Review)*. http://dx.doi.org/10.1029/2024JB029836

Plenkers, K., Manthei, G., & Kwiatek, G. (2022). Underground in-situ acoustic emission in study of rock stability and earthquake physics. In *Springer Tracts in Civil Engineering* (pp. 403–476). Springer Science; Business Media Deutschland GmbH. https://doi.org/10.1007/978-3-030-67936-1_16

Plenkers, K., Reinicke, A., Obermann, A., Doonechaly, N. G., Krietsch, H., Fechner, T., Hertrich, M., Kontar, K., Maurer, H., Philipp, J., Rinderknecht, B., Volksdorf, M., Giardini, D., & Wiemer, S. (2023). Multi-disciplinary monitoring networks for mesoscale underground experiments: Advances in the bedretto reservoir project. *Sensors*, *23*. https://doi.org/10.3390/s23063315

Rosskopf, M., Durand, V., Villiger, L., Doetsch, J., Obermann, A., & Krischer, L. (2024). *DUGseis processing example with needed data files*. Zenodo. https://doi.org/10.5281/zenodo.10598393

DRAFT

Villiger, L., Gischig, V., Doetsch, J., Krietsch, H., Dutler, N., Jalali, M., Valley, B., Selvadurai, P., Mignan, A., Plenkers, K., & others. (2020). Influence of reservoir geology on seismic response during decameter-scale hydraulic stimulations in crystalline rock. *Solid Earth*, *11*, 627–655. https://doi.org/10.5194/se-11-627-2020

Villiger, Linus, Haag, T., & Doetsch, J. (2024). *DugSeis acquisition*. ETH Research Collection. https://doi.org/10.5281/zenodo.10728558

DRAFT